

The Carbon Bathtub

Separating a stock from a flow, and working out what it actually takes to stop a level rising.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/carbon-bathtub/

1 Before you open anything

Answer from what you already know. Getting it right is not the point — having something on paper to argue with later is.

PREDICTION

Global CO₂ emissions are cut by 30% this year and then held at that lower rate, unchanged, for the next fifty years. Over those fifty years, does the concentration of CO₂ in the atmosphere rise, fall, or stay level? By roughly how much?

Write one sentence saying what reasoning got you there.

2 Read the tap, the drain and the level

Open the demonstration. The segmented control at the top of the stage should be on **Carbon**. Nothing in this part runs the clock — you are only moving the tap and reading what the panels say.

1. Press **Reset**. The **Year** box should read 2025, the **Emissions** slider 41.0 Gt CO₂ yr⁻¹, and **Stock — atmospheric CO₂** should read 424.0 ppm.
2. Fill in the first row of the table from the panels on the right. Do *not* press Run.
3. Press the **Cut 30%** preset button and fill the second row. Then **Cut 50%**, then **Net zero**.

Preset	Inflow – emissions (Gt CO ₂ yr ⁻¹)	Outflow – ocean + land uptake (Gt CO ₂ yr ⁻¹)	Level is changing at (ppm yr ⁻¹)	Emissions that would hold the level steady	Where the level ends up (ppm)
Today, 41					
Cut 30%					
Cut 50%					
Net zero					

a. Two whole columns hold the same value in all four rows. Name them. You changed the tap four times, so those numbers cannot be controlled by the tap – what are they controlled by instead?

b. The **Stock – atmospheric CO₂** box also read 424.0 ppm every time. Explain why moving the emissions slider changed how fast the level moves without changing the level itself.

c. Inflow and outflow are quoted in Gt CO₂ per year; the stock is quoted in ppm. Read the caption under the bathtub and write down the single conversion factor it uses to get from one to the other.

The two quantities are not interchangeable. A **flow** is measured per unit time and a **stock** is not, which is why they take different units and why “emissions” and “concentration” can move in opposite directions in the same year.

3 Now run the clock

One control, four settings, fifty simulated years each. This is the part that settles your prediction.

1. Press **Reset**. This returns the year to 2025, the concentration to 424.0 ppm *and* the emissions slider to 41.0 – so set the preset *after* resetting, never before.
2. Leave emissions at **41.0**. Press **Skip 10 years** five times, until the **Year** box reads 2075.
3. Record the stock and the two right-hand readouts.
4. Press Reset, choose the next preset, and repeat for all four rows.

Emissions held (Gt CO ₂ yr ⁻¹)	CO ₂ in 2025 (ppm)	CO ₂ in 2075 (ppm)	Change over 50 yr (ppm)	Outflow in 2075 (Gt CO ₂ yr ⁻¹)	Level is changing at, 2075 (ppm yr ⁻¹)
41.0 – Today					
28.7 – Cut 30%					
20.5 – Cut 50%					
0 – Net zero					

a. One of your rows holds emissions 30% below today's figure – a very large cut, sustained for half a century – and the concentration climbs anyway. Write the rule that decides whether the level rises, in one sentence, using the words *inflow* and *outflow*.

b. Go back to your part-1 prediction. Compare it with the "Cut 30%" row. If it did not match, name the specific assumption you made that turned out not to hold.

c. In the top two rows, the outflow in 2075 is larger than the 22.3 you recorded in part 2 – and there is no control anywhere in the demonstration that sets the drain. What made the outflow grow?

d. Two rows show the concentration falling. Using the outflow you recorded in part 2, state the condition those two satisfy in 2025 and the other two do not.

4 Working backwards

So far every step handed you an emissions rate and asked what happened. Questions usually run the other way: here is the outcome, find the rate.

1. Using your part-3 table, take any two rows and compute **outflow ÷ (CO₂ in 2075 – 280 ppm)**.

2. Press Reset. Drag the **Emissions** slider until **Where the level ends up if emissions stay exactly where the slider is** reads 500 ppm. The slider moves in steps of 0.1, so click it once and use the arrow keys for the last tenth. Record the emissions value.

3. Press Reset, leave emissions at 41.0, and press **Skip 10 years** five times to reach 2075. Read **Emissions that would hold the level steady right now** and the small line of text underneath it.

a. Step 1 gives you the same constant from both rows. Write it down with its units, then write the outflow as an equation in terms of the concentration C .

b. A level stops moving when outflow equals inflow. Use your equation to work out, on paper, the emissions rate that holds the concentration at 500 ppm. Compare it with the slider value you found in step 2.

c. In 2025 the demonstration said $22.3 \text{ Gt CO}_2 \text{ yr}^{-1}$ would hold the level steady, a cut of 45.6% from 41.0. In 2075, after fifty years at 41.0, it names a *larger* emissions figure and a *smaller* percentage cut. Explain why waiting makes the required cut look easier, and say what has been paid for that.

Read the **Honest limits** panel before you push part (c) any further. This model lets the sink keep growing without limit as concentration rises, whereas real ocean and land sinks saturate; the real carbon cycle also has several sinks on very different clocks and a tail that keeps a large fraction of added CO_2 in the air for centuries. Both simplifications make stabilisation look easier here than it is.

5 Solve these without the demonstration

Close the page or look away from it. Show your working. Then reopen it, check yourself, and if a number is off, write down where the arithmetic went astray.

Three quantities are all you need, and you have already met each one: pre-industrial concentration **280 ppm**; the uptake constant from part 4, **$0.155 \text{ Gt CO}_2 \text{ yr}^{-1}$ per ppm of excess above 280**; and **7.79 Gt CO_2 per ppm**. Every problem below sits at the 2025 state, 424 ppm, so you can check each one by pressing Reset and moving the slider only.

1. The concentration is 424 ppm and emissions are held at $35.0 \text{ Gt CO}_2 \text{ yr}^{-1}$. Find the outflow, the net flow, and the rate at which the concentration is changing in ppm yr^{-1} . State whether the level is rising or falling.

2. Emissions are to be held constant at whatever rate eventually settles the concentration at 600 ppm. Find that rate. Then say how it compares with today's $41.0 \text{ Gt CO}_2 \text{ yr}^{-1}$, and what that comparison implies about a "600 ppm world".

3. A news report states that emissions fell from 41.0 to $33.0 \text{ Gt CO}_2 \text{ yr}^{-1}$ and describes this as "a reduction in atmospheric carbon dioxide." Starting from 424 ppm, calculate the rate of change of the concentration at 33.0. Then state precisely which part of the report is correct and which part is not.

4. No calculation for this one. Explain why the emissions rate that holds the concentration steady is not a fixed number, but climbs as the concentration climbs. Then state what would have to become true of the ocean and land sinks for that to stop being the case.

6 On your own

Nothing here is graded, and none of it was needed above. The demonstration carries a second system that runs on the identical arithmetic.

- Switch the stage control to **Runoff**. Set *Heavy, 6.0*, press Run at 3× and watch dissolved oxygen fall past the fish-kill line. Then press *Stop the fertiliser* and keep running. The tap is off and nothing new is entering — so read the box that says how many days oxygen still needs, and find out which tub is responsible.
- In Runoff mode the four tabs — Nutrient, Algae, Dead matter, Oxygen — chart four separate bathtubs. After the fertiliser is stopped, put them in order from fastest-emptying to slowest. The captions under the chart quote a time constant for each; check whether your ordering agrees with them.
- On the lower chart panel in either mode, the shaded band between the two flow lines is exactly how fast the level above is moving. Find a setting where the band closes to nothing, and look at what the stock curve is doing at that moment. Then switch to *Cut 50%* and see which side of the other line the band moves to.